

Problem 2

The speed of a car during the first half of the distance is $v_1 = 36$ km/h, and during the second half of the distance is $v_2 = 54$ km/h. What is the average speed of the car over the entire journey?

Solution

To calculate the average speed when the distances are equal, we use the formula:

$$v_{\text{avg}} = \frac{2v_1v_2}{v_1 + v_2}.$$

Given:

$$v_1 = 36 \text{ km/h}, \quad v_2 = 54 \text{ km/h},$$

Substituting the values:

$$v_{\text{avg}} = \frac{2 \cdot 36 \cdot 54}{36 + 54}.$$

Simplify:

$$v_{\text{avg}} = \frac{2 \cdot 1944}{90} = \frac{3888}{90} = 43.2 \text{ km/h}.$$

Thus, the average speed of the car over the journey is:

$$v_{\text{avg}} = 43.2 \text{ km/h}.$$